YOUTUBE CHANNEL
S KHAN ACADEMY

Date:
28-02-23 T: 5:00pm

MTH643

Q1: $\int_0^2 e^{x^2} dx$, $n = 20$ (Rieman function)

Q2: i) Augmented matrix ii) Reduce echelon form

$$2x + y + 2 = 2$$
$$-x + y - z = 3$$
$$x + 2y + 3z = -10$$

Q3: $\int_1^2 (\cos(x) - \ln(x) + e^x) dx$, $n = 10$ through simpson rule.

Q4: $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = \sin x$, $y(0) = 1$, $y'(0) = 0$ through diff. eq.

Q5: $\int_0^{\pi} \sin(x) dx$, $n = 6$ through trapezoidal rule.

Q6: Ak Q function or circle calculation may area or circumference ki output find kerna thi.

Q7: $\sum_{x=1}^{n} \left(\frac{1}{x}\right)^x$, $n = 5$ Wala Q.

Q8: $y' = 2t$, $y(0) = 6$ ← Initial condition, $[0, 5]$ → interval
find diff. eq.

Q9: Mcq's basic concept kay thay zaida functions ko denote kerna tha. App ne may saray functions ki short form zaroor identify kerni hay

MATLAB Drive > untitled90.m

```
1  %Question No 1 reimann sum
2
3  clc
4  close all
5  clear all
6  f=@(x)exp(x^2);
7  a=0;
8  b=2;
9  n=20;
10 dx=(b-a)/n;
11 sum=0;
12 for i=1:n;
13 c=a+i*dx;
14 sum=sum+f(c);
15 end
16 Area=sum*dx
```

S KHAN ACADEMY

YOUTUBE CHANNEL
S KHAN ACADEMY

```
Area =

   19.3139


>>
```

S KHAN ACADEMY

MATLAB Drive > untitled91.m*

```
1  %Question no 2
2
3  clc
4  clear all
5  close all
6  A=[2 1 1;-1 1 -1;1 2 3];
7  b=[2;3;-10];
8  %Arument matrix
9  Ab=[A b]
10 % row reduce echolen form
11 D=rref(Ab)
12
```

S KHAN ACADEMY

```
Ab =

     2     1     1     2
    -1     1    -1     3
     1     2     3   -10

D =

     1     0     0     3
     0     1     0     1
     0     0     1    -5

>>
```

S KHAN ACADEMY

YOUTUBE CHANNEL
S KHAN ACADEMY

MATLAB Drive > untitled92.m

```
%Question no 3

%simpson1/3
clc
clear all
close all
f=@(x)(cos(x)-log(x)+exp(x));
a=1;
b=2;
n=100;
h=(b-a)/n;
sum1=0;
sum2=0;
for k=1:n-1
x(k)=a+k*h;
if mod(k,2)==0
 sum1=sum1+f(x(k));
else
 sum2=sum2+f(x(k));
end
end
answer=h/3*(f(a)+f(b)+2*sum1+4*sum2)
```

S KHAN ACADEMY

```
answer =

    4.3523


>>
```

S KHAN ACADEMY

MATLAB Drive > untitled93.m*

```
1  %Question no 4
2  clc
3  clear all
4  close all
5  syms y(x)
6  ode=diff(y,x,2)+2*diff(y,x)+5*y==sin(x);
7  dy=diff(y,x);
8  cond1=y(0)==1;
9  cond2=dy(0)==0;
10 cond=[cond1 cond2];
11 sol=dsolve(ode,cond)
```

S KHAN ACADEMY

```
sol =

cos(2*x)*(cos(3*x)/40 + (3*sin(3*x))/
40 - cos(x)/8 - sin(x)/8) -
sin(2*x)*((3*cos(3*x))/40 -
sin(3*x)/40 - cos(x)/8 + sin(x)/
8) + (11*cos(2*x)*exp(-x))/10 +
(9*sin(2*x)*exp(-x))/20


>>
```

S KHAN ACADEMY

YOUTUBE CHANNEL
S KHAN ACADEMY

MATLAB Drive > untitled94.m

```
%Question no 5

%trapozidl rule
clc
clear all
close all
f=@(x)(sin(x));
a=0;
b=pi;
n=10;
h=(b-a)/n;
sum=0;
for i=1:n-1
 x(i)=a+i*h;
 sum=sum+f(x(i));
end
answer=h/2*(f(a)+f(b)+2*sum)
```

S KHAN ACADEMY

```
answer =

    1.9835


>>
```

S KHAN ACADEMY

MATLAB Drive > myfunction.m

```
1 function[Area,Circum]=myfunction(r)
2 Area=pi*r^2
3 Circumference=2*pi*r
4 end
5 % run krain or phir function ko commond
6 %main call krain
```

S KHAN ACADEMY

```
>> myfunction(4)

Area =

    50.2655

Circumference =

    25.1327

ans =

    50.2655

>>
```

S KHAN ACADEMY

MATLAB Drive > untitled96.m

```
1  %Question No 7
2
3  clc
4  clear all
5  close all
6  syms x
7  n=5;
8  sum=0;
9  for k=1:n
10 sum=sum+(3/x)^k;
11 end
12 disp(sum)
```

S KHAN ACADEMY

```
3/x + 9/x^2 + 27/x^3 + 81/x^4 + 243/x^5

>>
```

S KHAN ACADEMY

MATLAB Drive > untitled97.m

```
1  %Question no 8
2
3  clc
4  clear all
5  close all
6  f=@(t,y)2*t;
7  interval=[0,5];
8  y0=0;
9  [t,y]=ode45(f,interval,y0)
10 plot(t,y)
```

S KHAN ACADEMY

YOUTUBE CHANNEL
S KHAN ACADEMY

```
t =

         0
    0.1250
    0.2500
    0.3750
    0.5000
    0.6250
    0.7500
    0.8750
    1.0000
    1.1250
    1.2500
    1.3750
    1.5000
    1.6250
    1.7500
    1.8750
    2.0000
    2.1250
    2.2500
    2.3750
    2.5000
    2.6250
    2.7500
    2.8750
    3.0000
    3.1250
    3.2500
    3.3750
    3.5000
    3.6250
    3.7500
    3.8750
    4.0000
    4.1250
    4.2500

>>
```

S KHAN ACADEMY

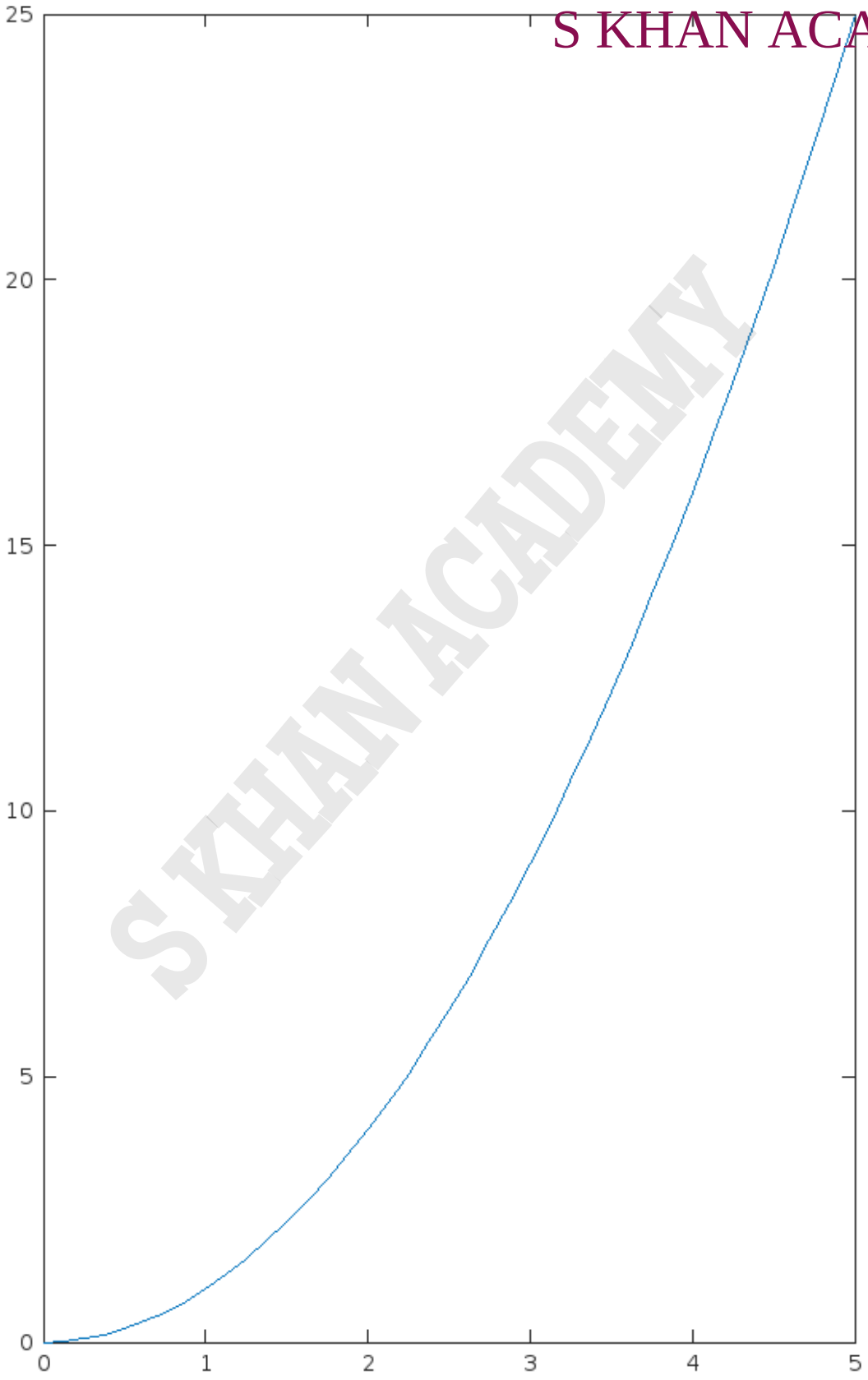
YOUTUBE CHANNEL
S KHAN ACADEMY

```
y =

         0
    0.0156
    0.0625
    0.1406
    0.2500
    0.3906
    0.5625
    0.7656
    1.0000
    1.2656
    1.5625
    1.8906
    2.2500
    2.6406
    3.0625
    3.5156
    4.0000
    4.5156
    5.0625
    5.6406
    6.2500
    6.8906
    7.5625
    8.2656
    9.0000
    9.7656
   10.5625
   11.3906
   12.2500
   13.1406
   14.0625
   15.0156
   16.0000
   17.0156
   18.0625
   19.1406
```

S KHAN ACADEMY

YOUTUBE CHANNEL
S KHAN ACADEMY



S KHAN ACADEMY

SANAULLAH KHAN (Msc Mathematics)
For Online Tuision, Paid Quiz & Assignment Whatsapp (03066793776)